Multiple regression and conclusions

Overview

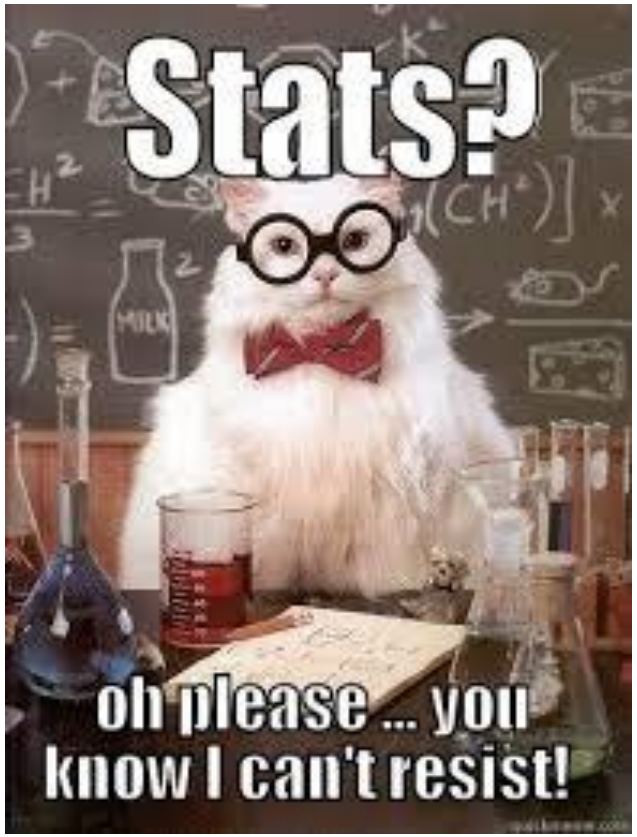
Review of theories of hypothesis testing

Review and continuation of inference for linear regression

Quick discussion of multiple regression

How to determine which CI/hypothesis test to use

Conclusions



Announcement

Final exam review session

- Monday April 27th from 1 pm
- In this classroom (Marsh)

Final exam:

- May 5th (Tuesday) at 7pm
- Location: Marsh (this classroom)

Final project due Sunday April 26th

If you have accommodations, please reach out to SAS as soon as possible to schedule to take the exam with them

Announcement

No office hours on Friday

- I can hold additional office hours after the review session next Monday from 2:30-3:30 pm.

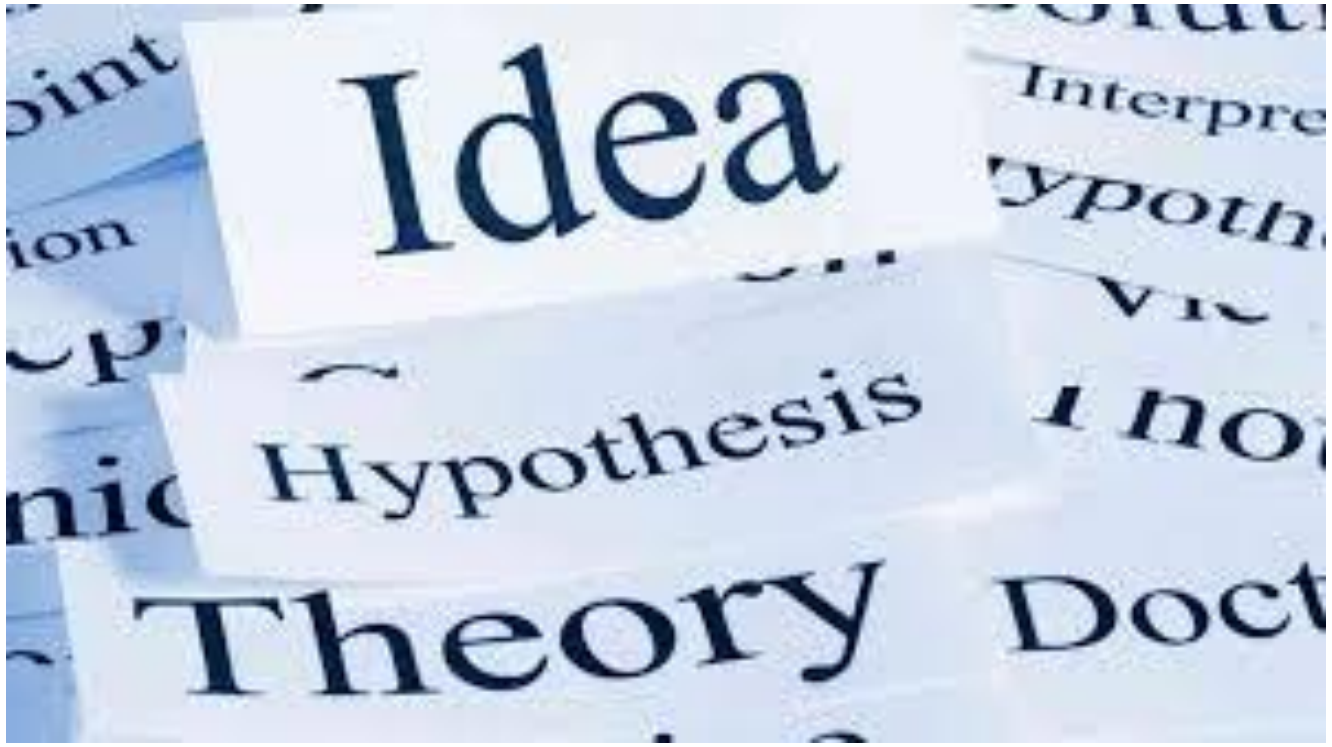
Download any data you want to save from the YCRC server

- Your account of the YCRC RStudio server will be deleted ~one month after the end of the semester
- You should be able to create a zip file of all your class material using:

```
Sys.setenv("R_ZIPCMD"="zip")

SDS1000::save_zip_of_all_class_material()
```

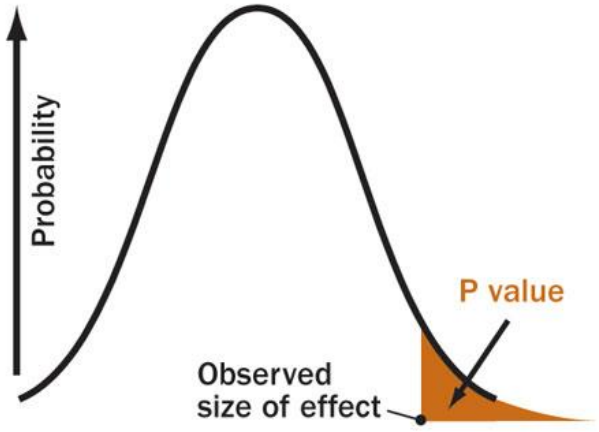
Quick review of theories of hypothesis tests



Two theories of hypothesis testing

1. **Significance testing** of Ronald Fisher
   - p-value as strength of evidence against the null hypothesis

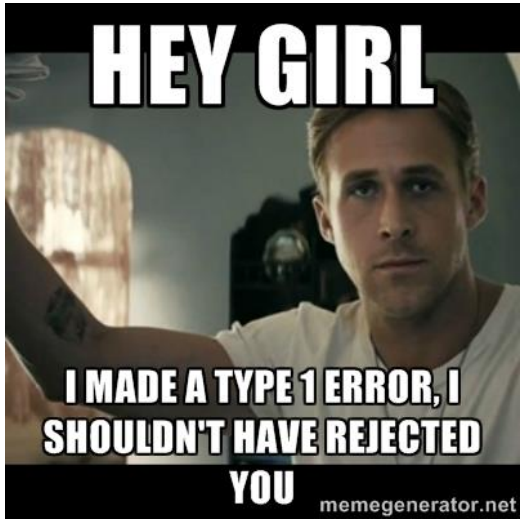
2. **Hypothesis testing** of Jezy Neyman and Egon Pearson
   - Make a formal decision of whether to reject H_0 (if p-value < predefined α value)

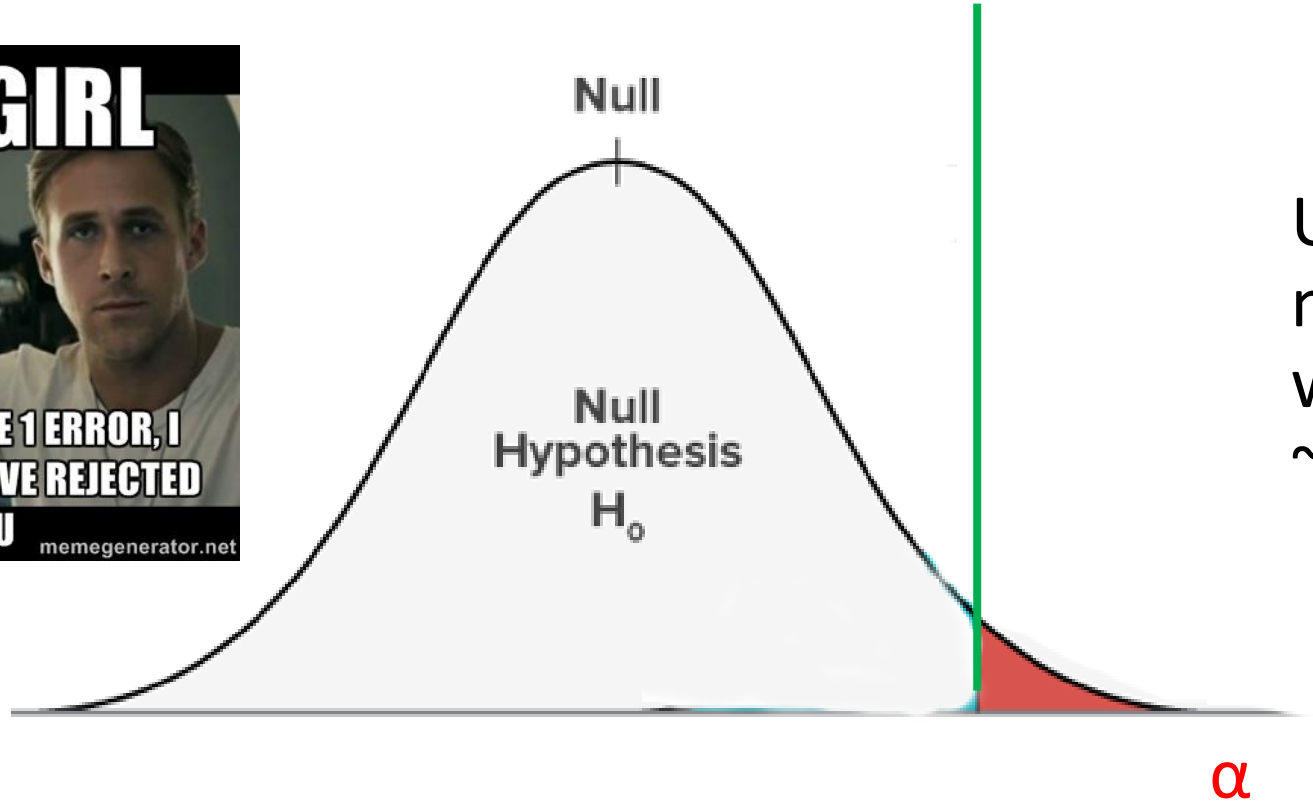


Significance testing

Hypothesis testing

Neyman-Pearson Frequentist logic

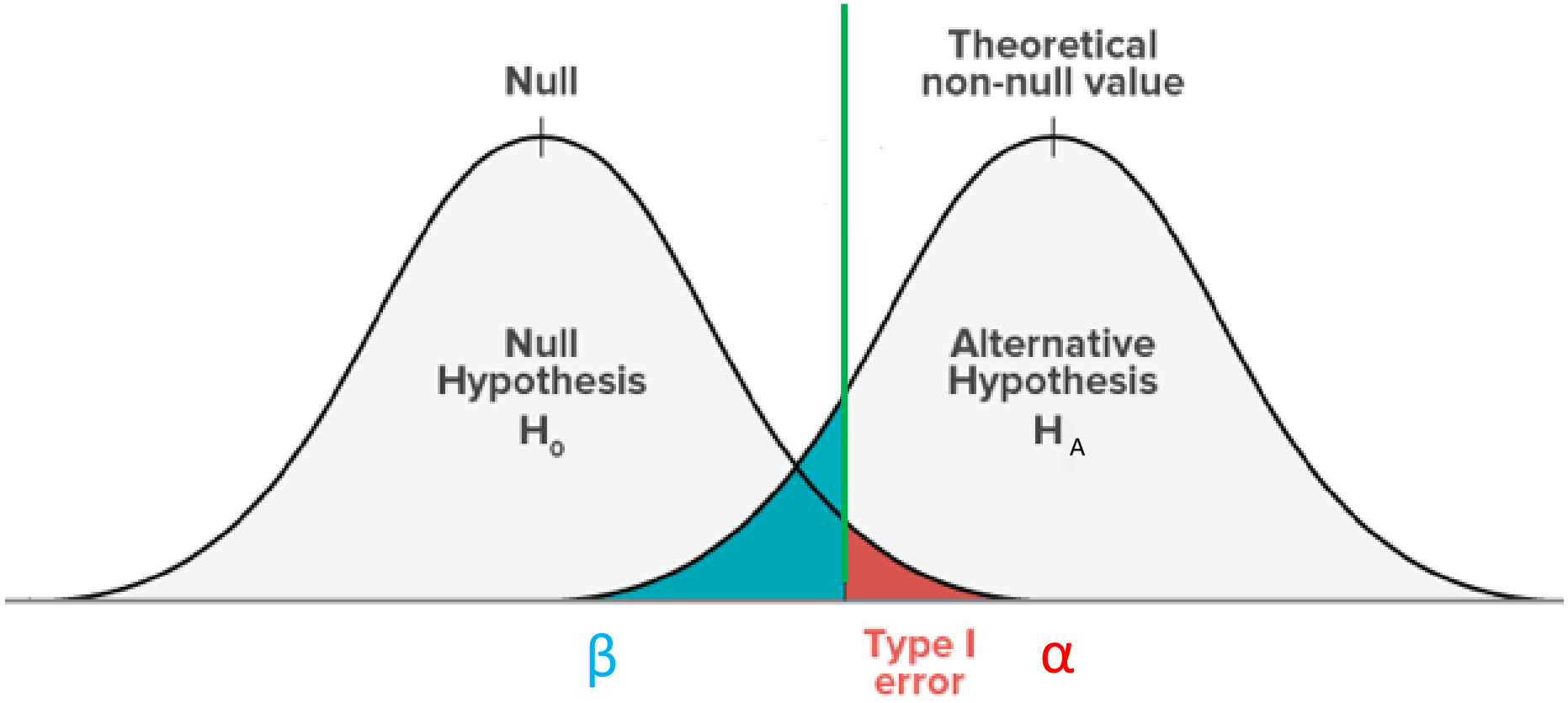




Using α = 0.05, when the null hypothesis is true, we would falsely reject it ~5% of the time

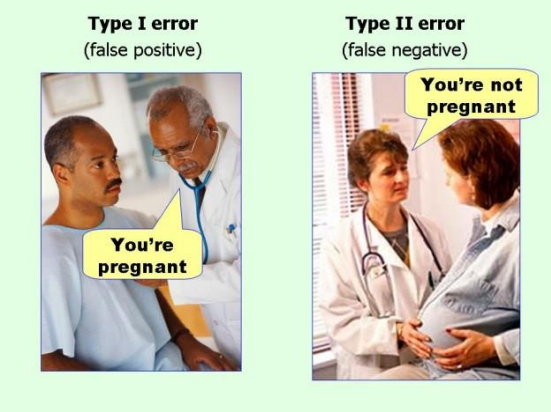
Type I error: incorrectly rejecting the null hypothesis when it is true

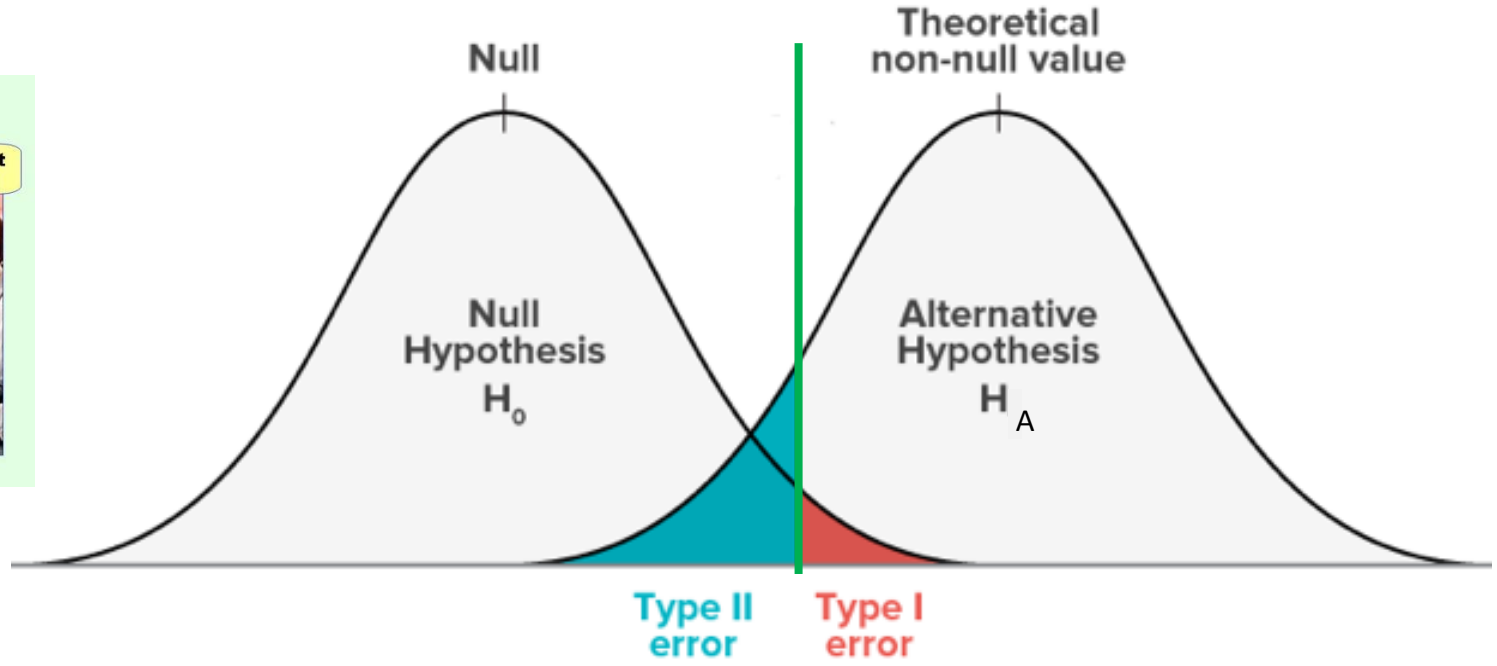
Neyman-Pearson Frequentist logic



Type II error: incorrectly rejecting failing to reject H_0 when it is false

Type I and Type II Errors





	H_0 is true	H_A is true (H_0 is false)
Reject H_0	Type I error (α) (false positive)	No error
Do not reject H_0	No error	Type II error (β) (false negative)

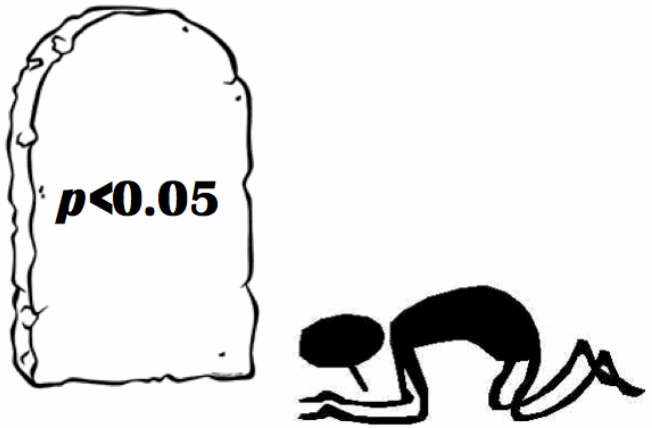
Problems with the NP hypothesis tests

<u>Problem 1</u>: we are interested in the results of a specific experiment, not whether we are right most of the time

- E.g., 95% of these statements are true:
  - Calcium is good for your heart, Paul is psychic, Buzz and Doris can communicate, …

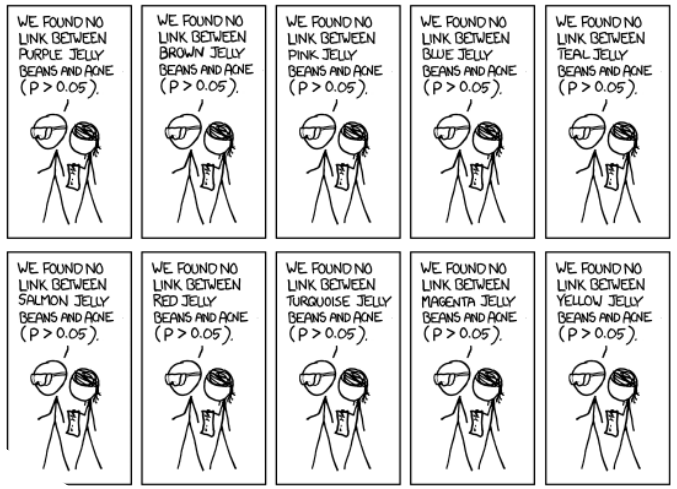
<u>Problem 2</u>: Arbitrary thresholds for alpha levels

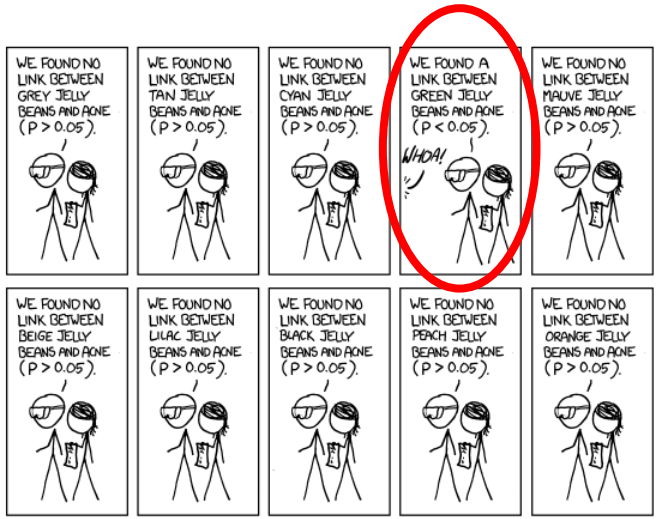
- P-value = 0.051, we don't reject H_0?



<u>Problem 3</u>: running many tests can give rise to a high number of type I errors

Multiple hypothesis tests





Don't ever do this!

Replication crisis

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

The file drawer effect

[American Statistical Association's ' Statement on p-values]

Some thoughts…

Better to have hypothesis tests than none at all. Just need to think carefully and use your judgment.

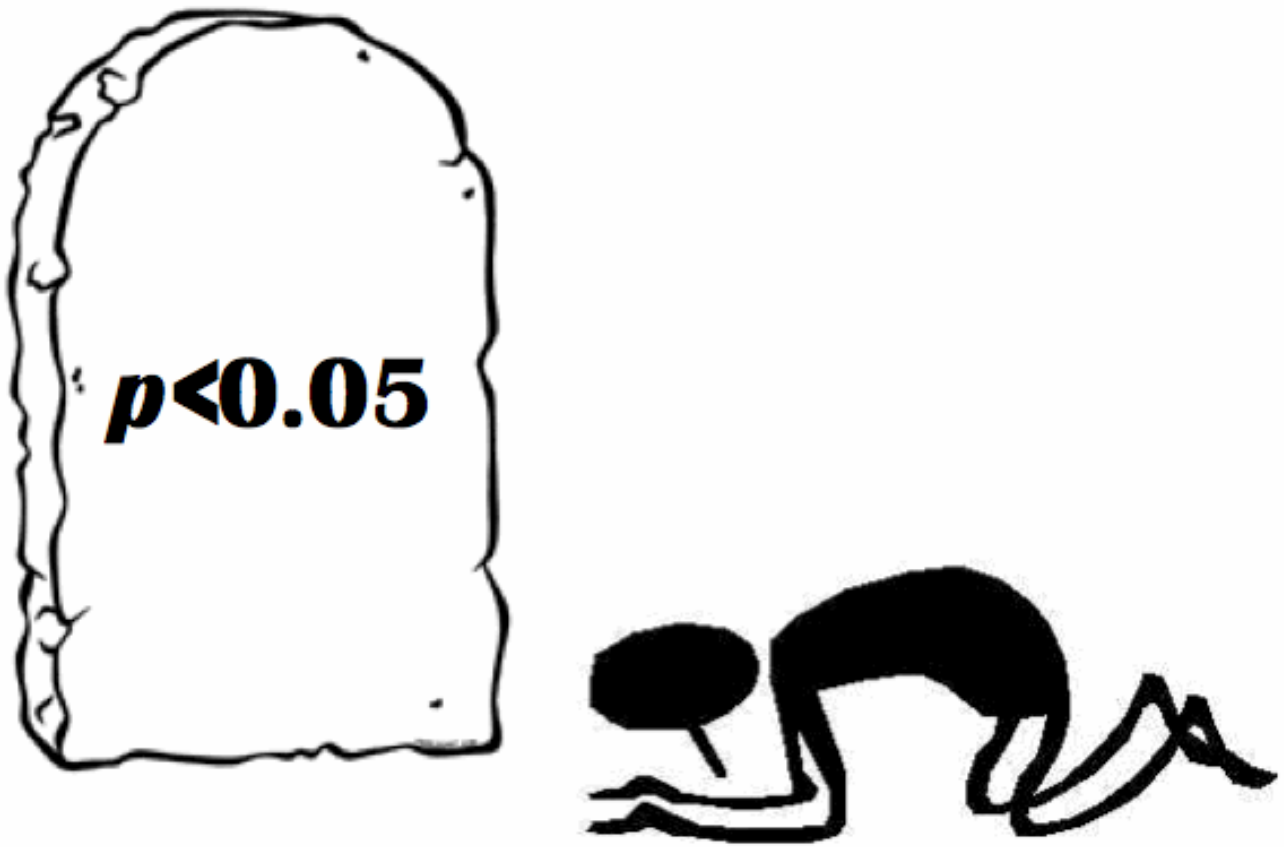
Report effect size in most cases – i.e., confidence intervals

Report the p-values rather than accept/reject H_0

- i.e., report $p = 0.023$ not $p < 0.05$

Replicate findings (perhaps in different contexts) to make sure you get the same results

Be a good/honest scientists and try to get at the Truth!


$p<0.05$

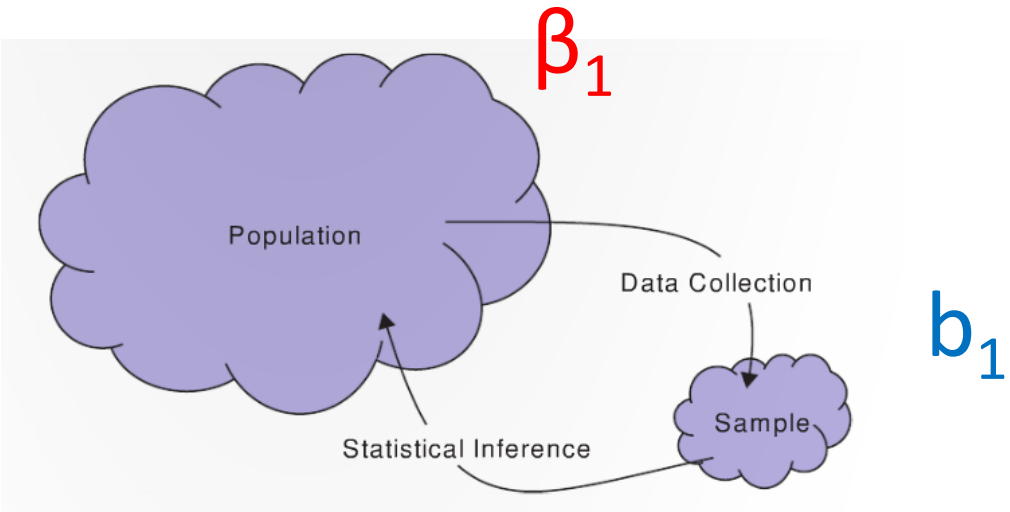
Review: Parametric inference for regression

Review of regression (class 6 and 7)

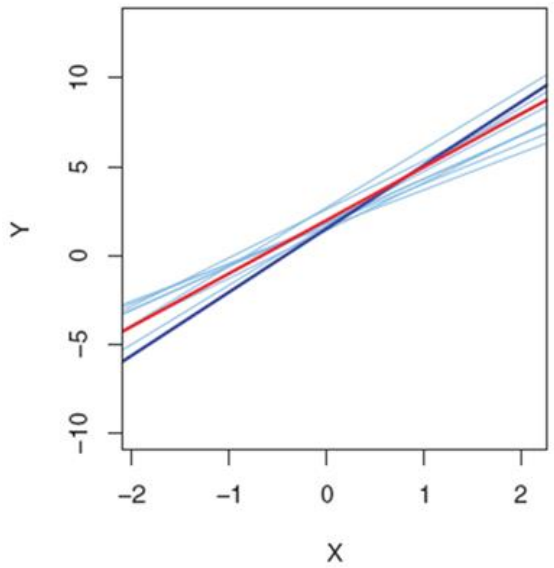
In **linear regression** we fit a <u>line</u> to the data, called the **regression line**

$$\hat{y} = b_0 + b_1 \cdot x$$

$$\textit{Predicted response} = b_0 + b_1 \cdot \textit{Explanatory}$$



b_1



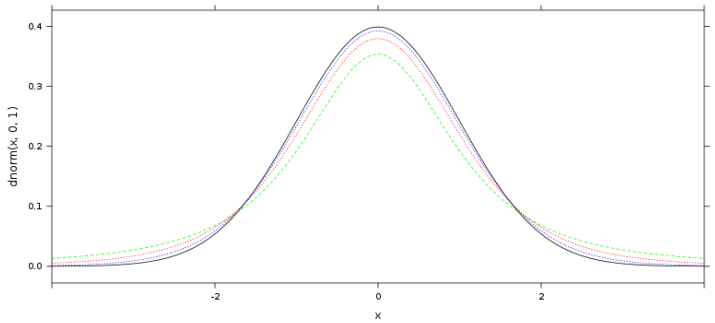
Hypothesis test for regression coefficients

We can run hypothesis tests to assess whether there is a linear relationship between y and x

- H_0: $\beta_1 = 0$ (slope is 0: no linear relationship between x and y
- H_A: $\beta_1 \neq 0$

One type of hypothesis test we can run is based on a t-statistic: $$t = \frac{b_1 - 0}{SE_{b_1}}$$

- The t-statistic comes from a t-distribution with n - 2 degrees of freedom

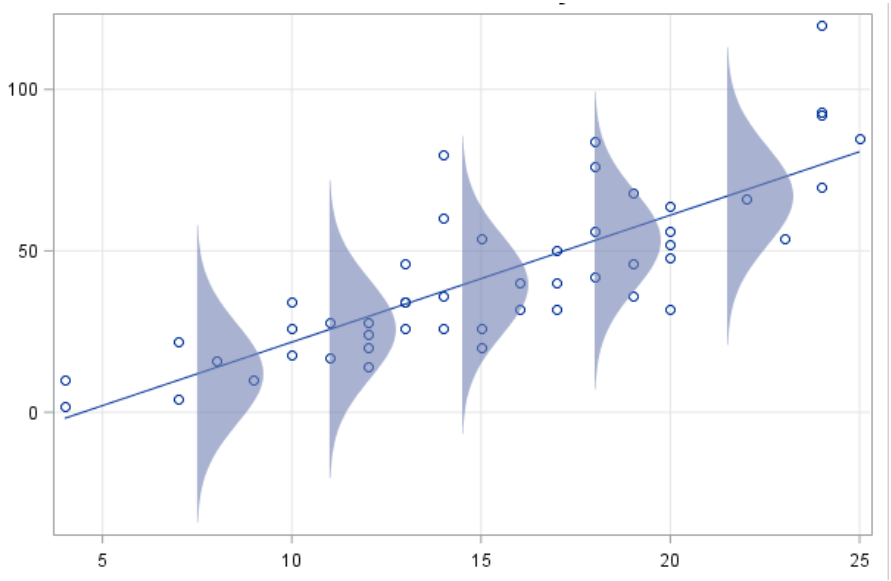


$$SE_{b_1} = \sqrt{\frac{1}{n-2}\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

Inference using parametric methods

When using parametric methods, we make the following (LINE) assumptions:

- **Linearity**: A line can describe the relationship between x and y
- **Independence**: each data point is independent from the other points
- **Normality**: errors are normally distributed
- **Equal variance (homoscedasticity)**: constant variance of errors over the whole range of x values



These assumptions are usually checked after the models are fit using 'regression diagnostic' plots.

Confidence intervals for regression coefficients

For the slope coefficient , the confidence interval is: $b_1 \pm t^* \cdot SE_{b_1}$

Where: $SE_{b_1} = \sqrt{\frac{1}{n-2}\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$

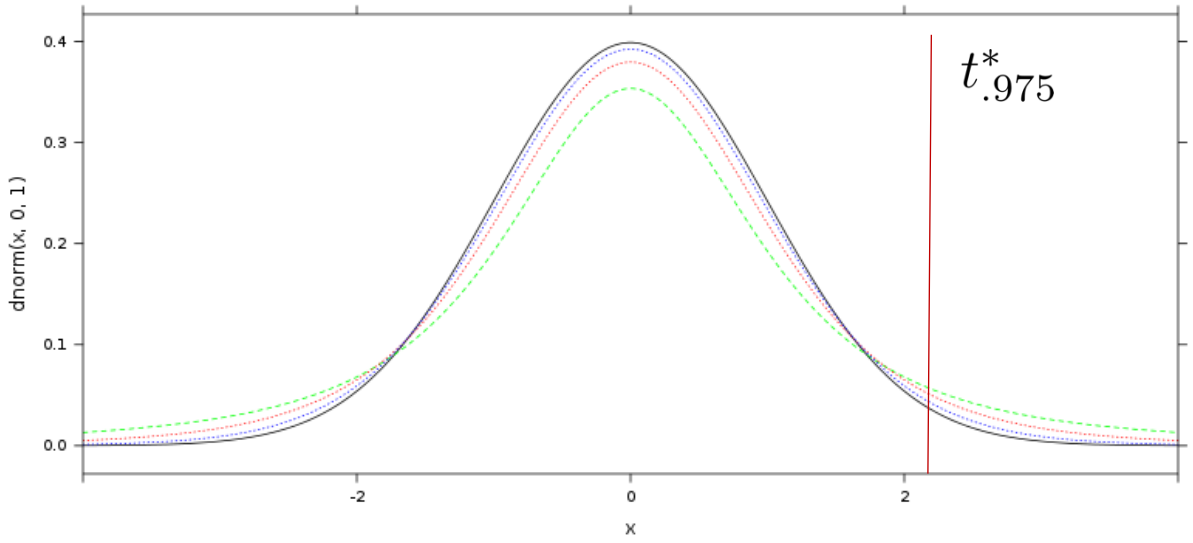
t* is the critical value for the t_{n-2} density curve needed to obtain a desired confidence level

N(0, 1)

df = 2

df = 5

df = 15



R code for parametric inference for regression

```
# fit our regression model

    lung_lm <- lm(cancer_rate ~ cigs_per_capita)



# run a hypothesis test for the regression coefficients

    summary(lung_lm)
```

Interpreting R output

b_0

$$t = \frac{b_1 - 0}{SE_{b_1}}$$

pt(t, df = n - 2)

```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     6.471686   2.140669   3.023  0.00425 **
cigs_per_capita 0.005291   0.000839   6.306 1.44e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b_1

$$SE_{b_1} = \sqrt{\frac{1}{n-2}\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

coef(lm_lung)

CI: $b_1 \pm t^* \cdot SE_{b_1}$

Multiple regression

Multiple regression

In multiple regression we try to predict a quantitative response variable *y* using several predictor variables $x_1, x_2, \ldots, x_k$

We estimate coefficients b_i using a data set to make predictions $\hat{y}$

$$\hat{y} = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k$$

The underlying population model is:

$$y = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \ldots \beta_k \cdot x_k + \epsilon$$

Multiple regression

$$\hat{y} = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + \ldots + b_k \cdot x_k$$

There are many uses for multiple regression models including:

- To make predictions as accurately as possible
- To understand which predictors (x) are related to the response variable (y)

Multiple regression

Let's predict first-year college GPA based data from 219 students using the following variables:

- High school GPA (HSGPA)
- Verbal SAT scores (SATV)
- Number of humanities credits (HU)

$$\hat{y}_{GPA} = b_0 + b_1 \cdot x_{HSGPA} + b_2 \cdot x_{SATV} + b_3 \cdot x_{HU}$$

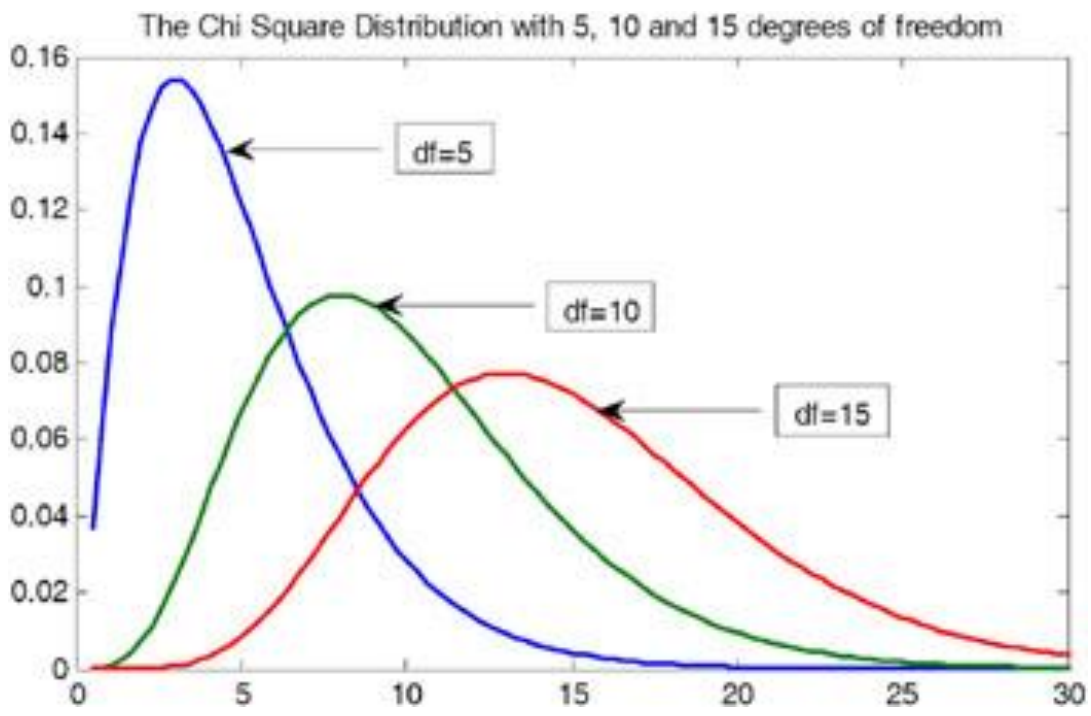
Let's quickly try it in R!

Choosing the appropriate hypothesis test and confidence interval

The parametric distributions

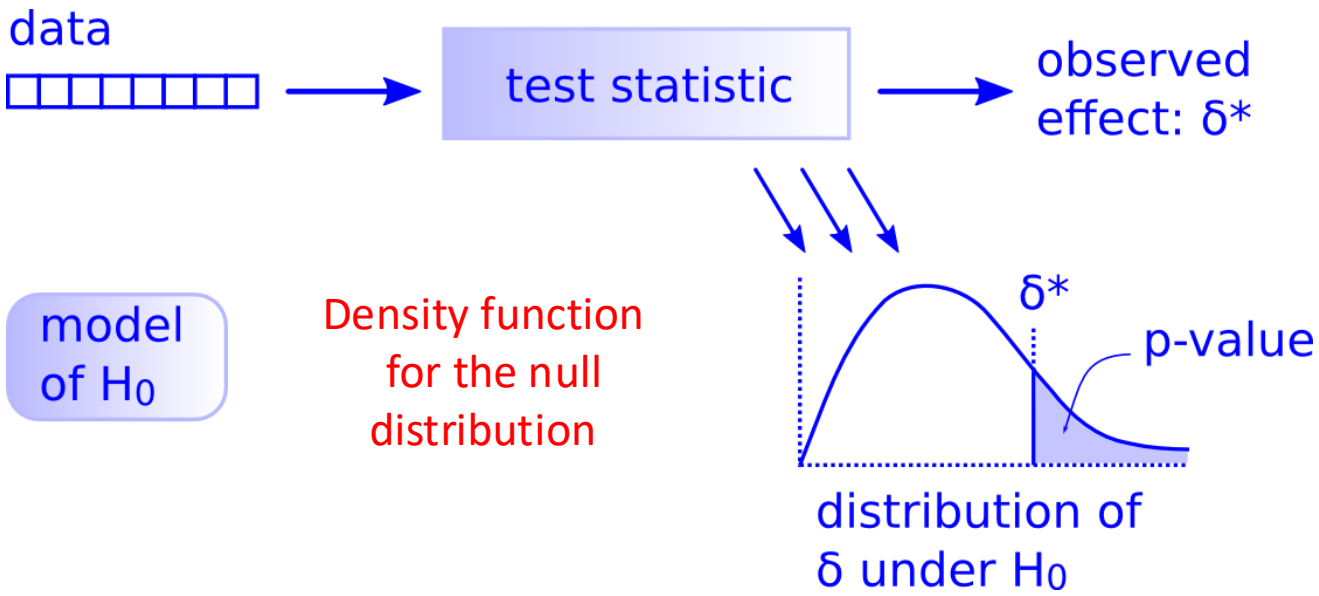
The parametric distributions we have discussed are:

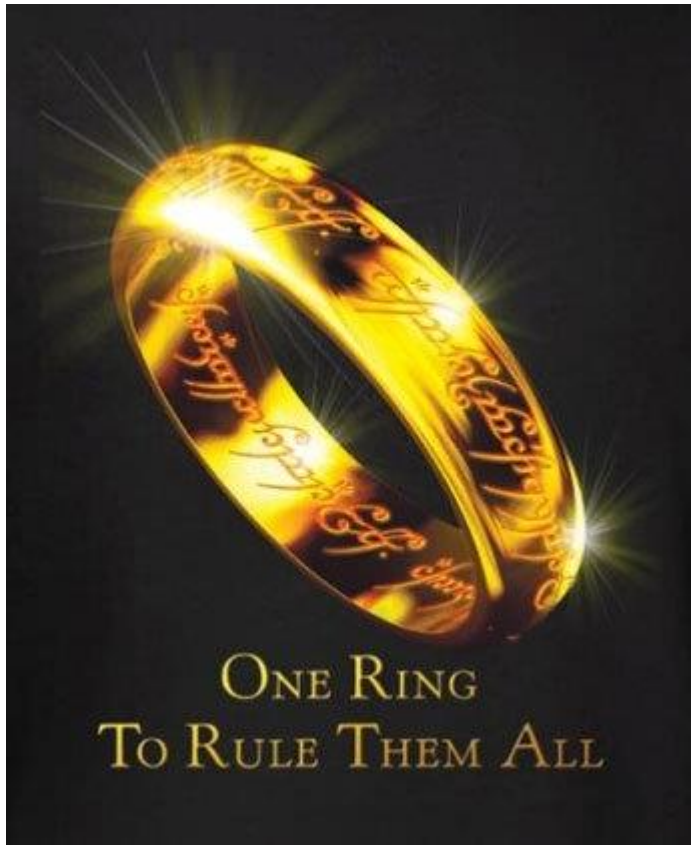
Name	Parameters
Normal	μ, $\boldsymbol{\sigma}$
t	df
χ^2	df
F	df_1, df_2



One test to rule them all

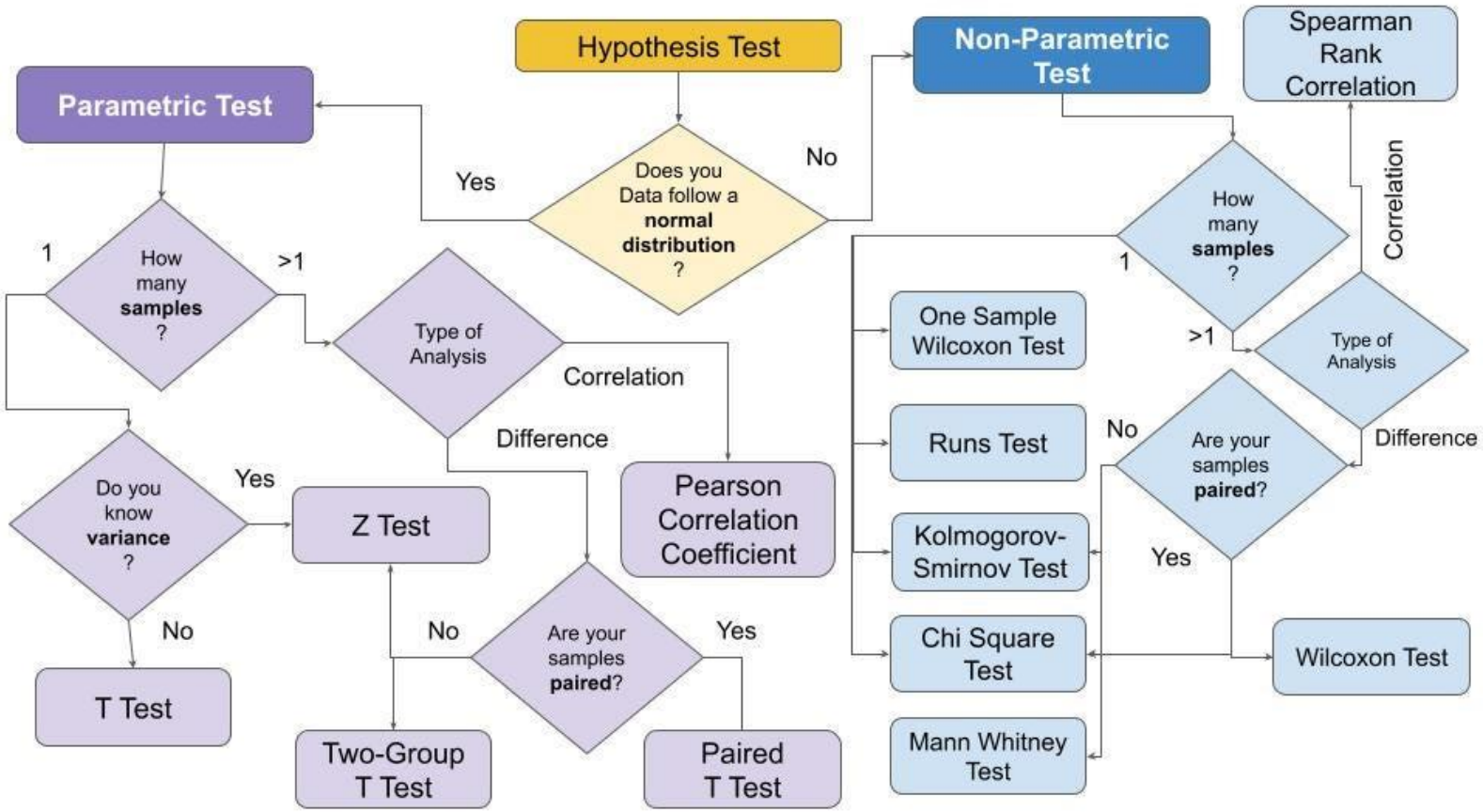
There is only one hypothesis test!





Just follow the 5 hypothesis tests steps!

Choosing the appropriate parametric test



Choosing the appropriate parametric test

Categorical data

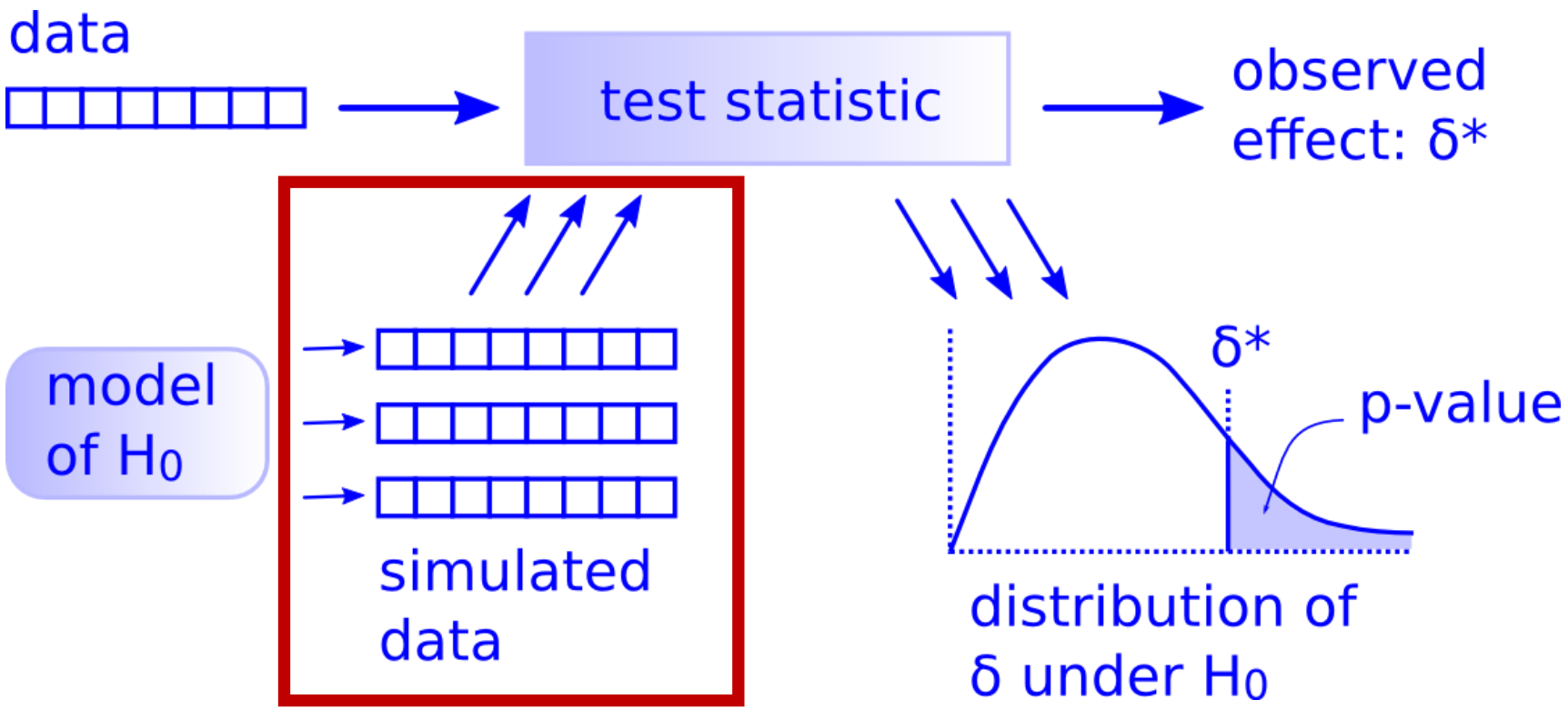
1 π	2 π	> 2 π
z-statistic		χ^2 statistic
pnorm()		pchisq()

Quantitative data

1 μ	2 μ	> 2 μ
t-statistic		F-statistic
pt()		pf()

Data	1 Sample	2 Samples	> 2 Samples
Categorical data	H_0: $\pi = p_0$ H_A: $\pi \neq p_0$ <u>z-test</u> $z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$	H_0: $\pi_1 = \pi_2$ H_A: $\pi_1 \neq \pi_2$ <u>z-test</u> $z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$	H_0: $\pi_1 = p_1$, $\pi_2 = p_2$, … , $\pi_k = p_k$ H_A: At least one p_i is different than specified <u>chi-square test</u> $\chi^2 = \sum_{i=1}^{k} \frac{(Observed_i - Expected_i)^2}{Expected_i}$ df = k - 1
Quantitative data	H_0: $\mu = v_0$ H_A: $\mu \neq v_0$ <u>One sample t-test</u> $t = \frac{\bar{x}-v_0}{s/\sqrt{n}}$ df = n - 1	H_0: $\mu_1 = \mu_2$ H_A: $\mu_1 \neq \mu_2$ <u>Two sample t-test</u> $t = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$ df = min n_1 - 1, n_2 - 1	H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$ H_A: At least one μ_i is different <u>Analysis of Variance</u> $F = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$ $df_1 = k - 1$, $df_2 = n - k$

Choosing the appropriate resampling method



Data	1 Sample	2 Samples	> 2 Samples
Categorical data	H_0: $\pi = p_0$ H_A: $\pi \neq p_0$ <u>Flip "coins"</u> rflip_count()	H_0: $\pi_1 = \pi_2$ H_A: $\pi_1 \neq \pi_2$ <u>Flip "coins"</u> rflip_count()	H_0: $\pi_1 = p_1$, $\pi_2 = p_2$, … , $\pi_k = p_k$ H_A: At least one p_i is different than specified <u>Roll a k-sided die n times</u> rroll(n, prob =)
Quantitative data	H_0: $\mu = v_0$ H_A: $\mu \neq v_0$ <u>resample</u> sample(… , replace = TRUE)	H_0: $\mu_1 = \mu_2$ H_A: $\mu_1 \neq \mu_2$ <u>Shuffle data</u> shuffle()	H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$ H_A: At least one μ_i is different <u>Shuffle data</u> shuffle()

Parametric confidence intervals

Confidence intervals have the form: $statistic \pm q^* \cdot SE$

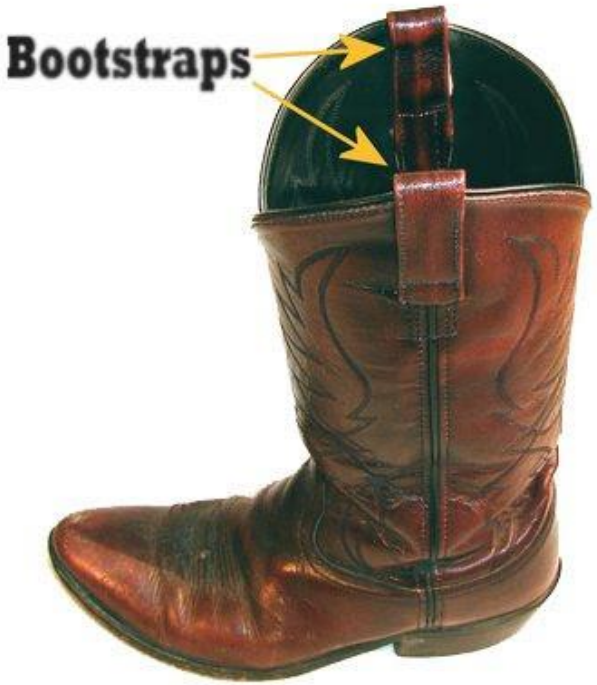
We just need the appropriate standard error (SE) formula

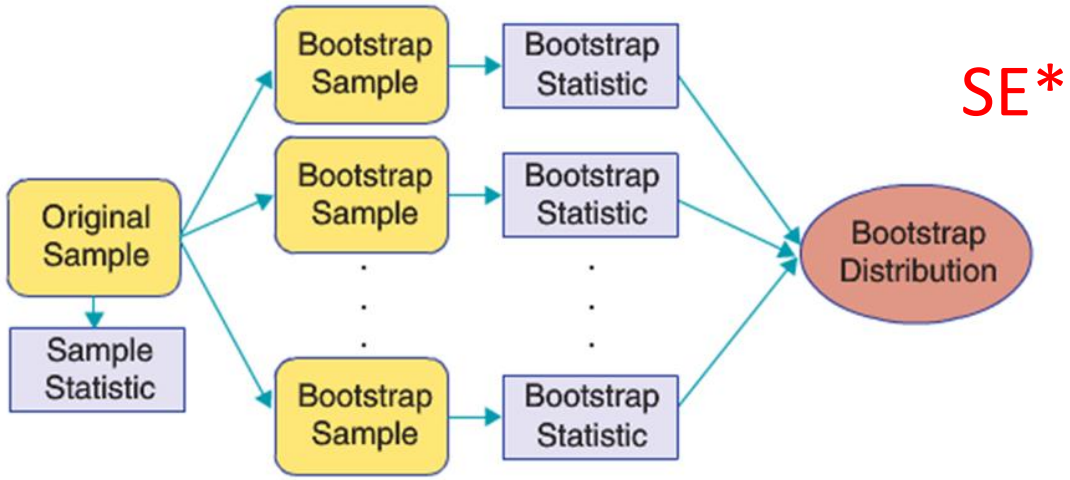
(and to determine if we should use t* or z*)

Data	1 Sample	2 Samples
Categorical Data	$SE = \sqrt{\frac{\pi(1-\pi)}{n}}$ $\hat{p} \pm z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$SE = \sqrt{\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}}$ $\hat{p_1} - \hat{p_2} \pm z^* \sqrt{\frac{\hat{p_1}(1-\hat{p_1})}{n_1} + \frac{\hat{p_2}(1-\hat{p_2})}{n_2}}$
Quantitative Data	$SE = \frac{s}{\sqrt{n}}$ $\overline{x} \pm t^* \frac{s}{\sqrt{n}}$	$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ $(\overline{x_1} - \overline{x_2}) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Computational confidence intervals

The bootstrap!





SE*

$$\textit{Statistic} \pm z^* \cdot SE^*$$

Additional hypothesis tests

Suppose in the future you want to test a hypothesis we have not covered in this class. What should you do?

- For example, H_0: $\sigma^2_1 = \sigma^2_2$

Write null and alternative hypotheses in symbols and then look up an appropriate test

- For example, H_0: $\sigma^2_1 = \sigma^2_2$ Levene's test, Bartlett's test, or the Brown–Forsythe test
- Make sure the conditions/assumptions for the test are met
  - See how robust the test is to violations to these assumption

Side note: **non-parametric** tests are another type of hypothesis test that makes fewer assumptions

- i.e., they do not assume that data comes from a normal distribution
  - Based on ranks, similar to the relationship between the mean and the median

How to use statistical methods to analyze real data

Know the scientific questions that you want to address and have data analysis plan ***before*** you collect data!

- i.e., state H_0's and H_A 's for the questions you want to address, find the appropriate test, etc.

Run a pilot study to get a sense of the data you will collect in your real study

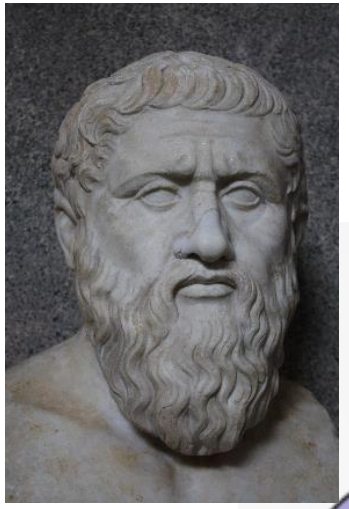
- Will give a sense of the distribution of the data (is the data normal, etc.)
- You can do power calculations as well to estimate the samples size n that you will need

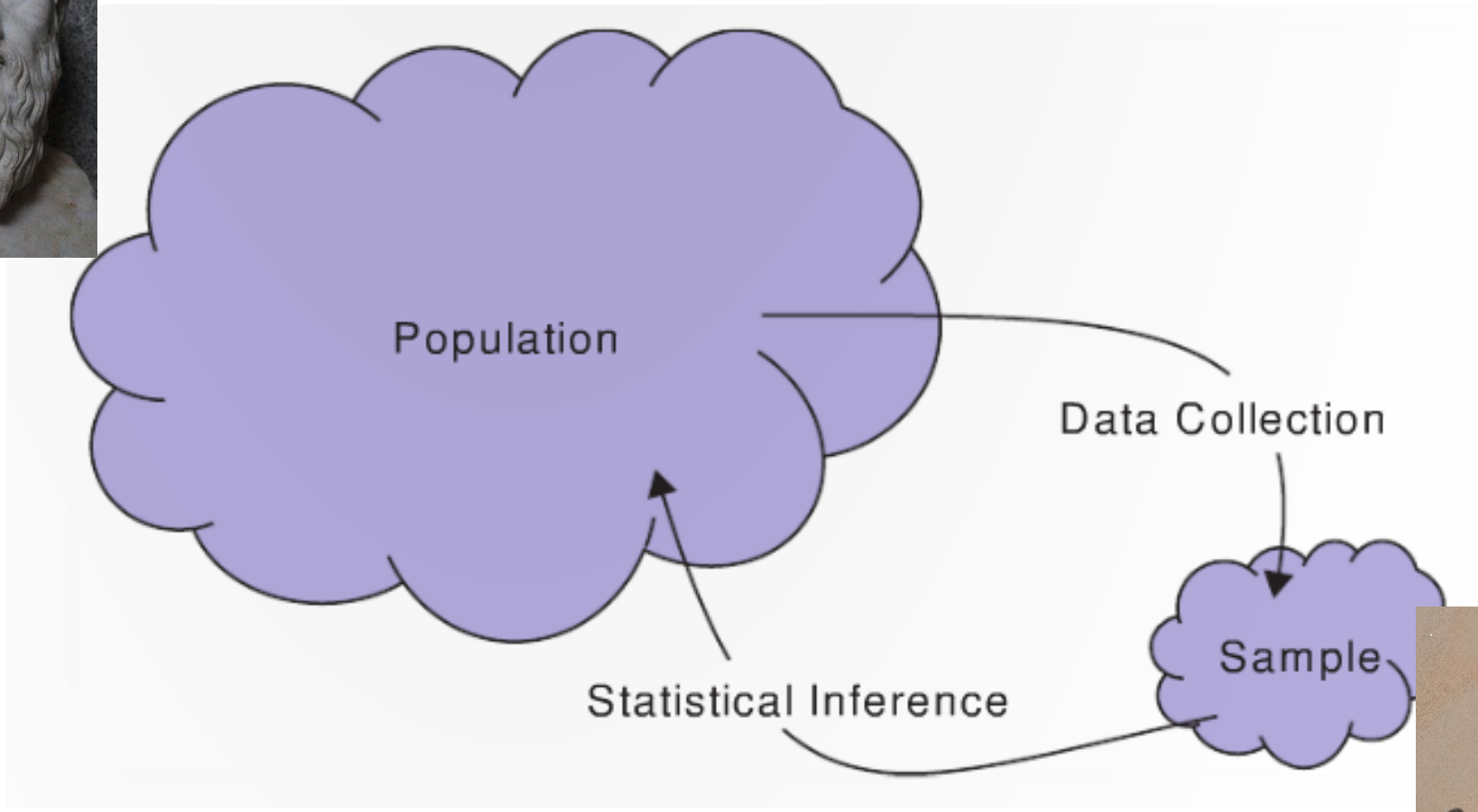
Ideally can pre-register your data analysis plan before collecting the data

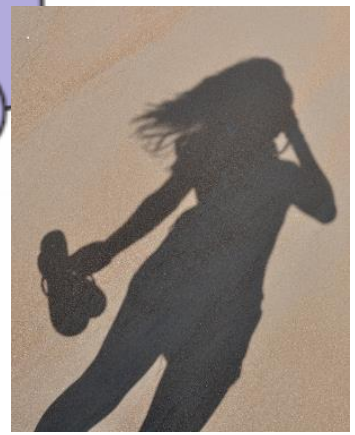
- Can help with the replication crisis

Conclusions

THE TRUTH IS OUT THERE




Population
Data Collection
Sample
Statistical Inference



Good luck studying for the finals and have a great summer!